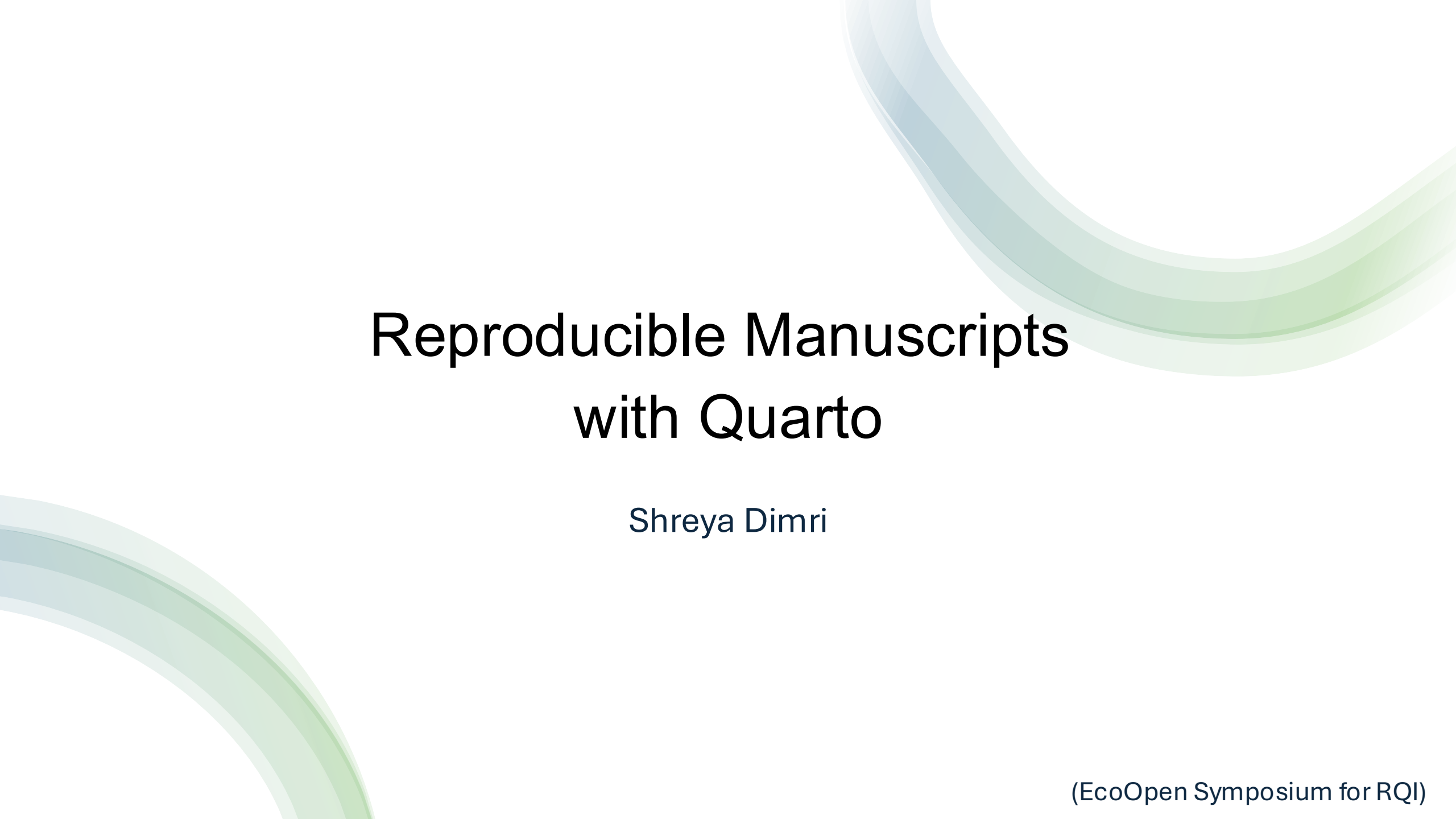


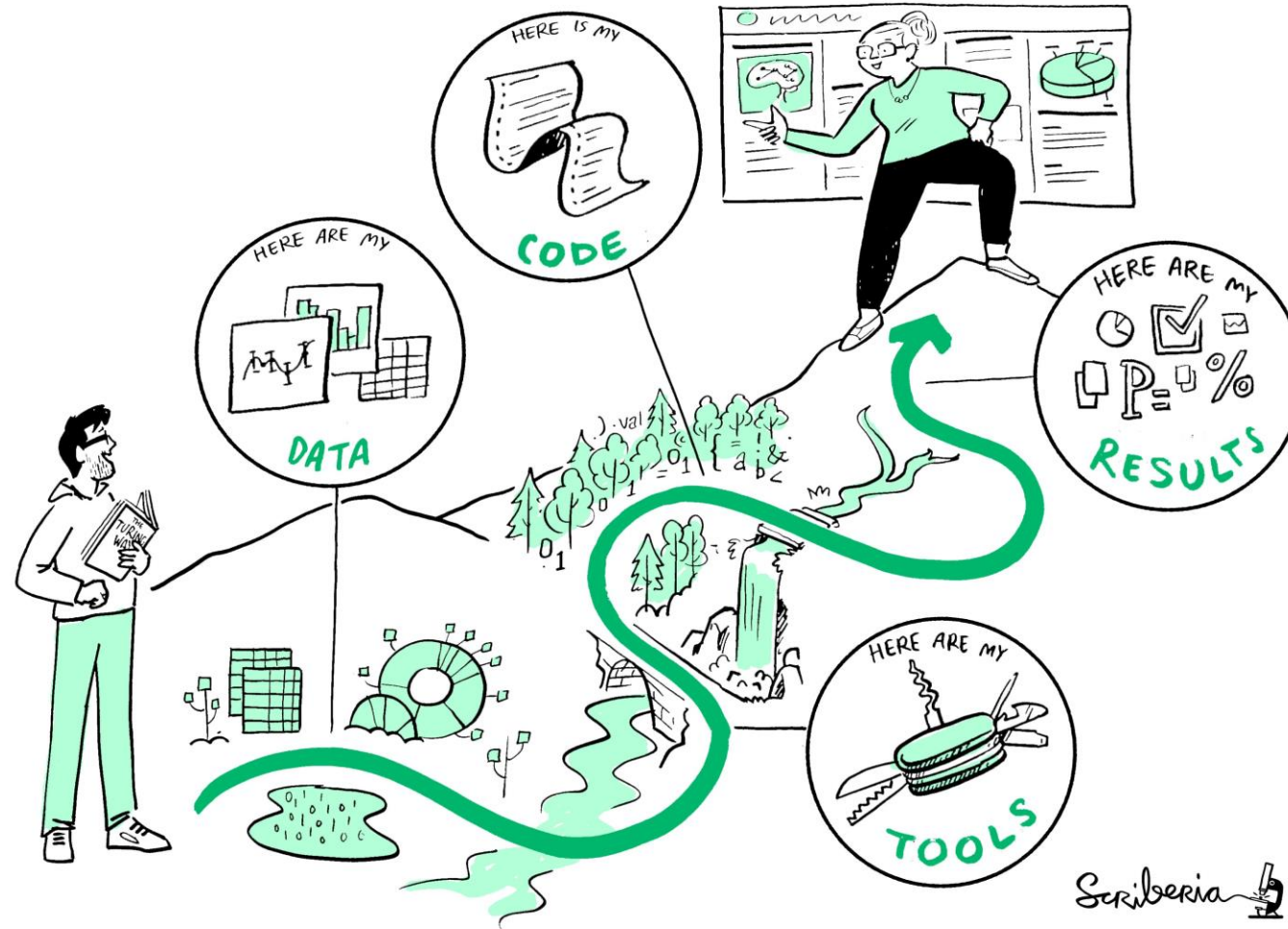


Reproducible Manuscripts with Quarto

Shreya Dimri

(EcoOpen Symposium for RQI)

What is meant by reproducible here?



But your research is often more than a PDF, it is a “knowledge stack”



Scriberia 

How do we write manuscripts at the moment?

- What do you use?
- And why?

How do we write manuscripts at the moment?

- What do you use?
- And why?
- The usual methods can have serious drawbacks i.e. barriers to reproducibility
- Re-submission cycle with painstaking reformatting

What do we want?

Dynamic Document Generation, where we can:

- combine narrative with code

What do we want?

Dynamic Document Generation, where we can:

- combine narrative with code
- automatically generate figures and tables
- automatically render results in text
- format the content into a scientific paper (including citations!)

What do we want?

Dynamic Document Generation, where we can:

- combine narrative with code
- automatically generate figures and tables
- automatically render results in text
- format the content into a scientific paper (including citations!)
- Repeat it whenever needed!

Benefits of a dynamic document

Eliminate human error in copying and pasting results

“We found that half of all published psychology papers that use null-hypothesis significance testing (NHST) contained at least one p-value that was inconsistent with its test statistic and degrees of freedom. One in eight papers contained a grossly inconsistent p-value that may have affected the statistical conclusion”

Benefits of a dynamic document

Easy revisions and specification of desired figures and tables

Promote computational reproducibility

- Easy verification and replication of research findings
- Ultimately reduce mistakes and maybe even save time (once you climb the mount learning)

What is Quarto?

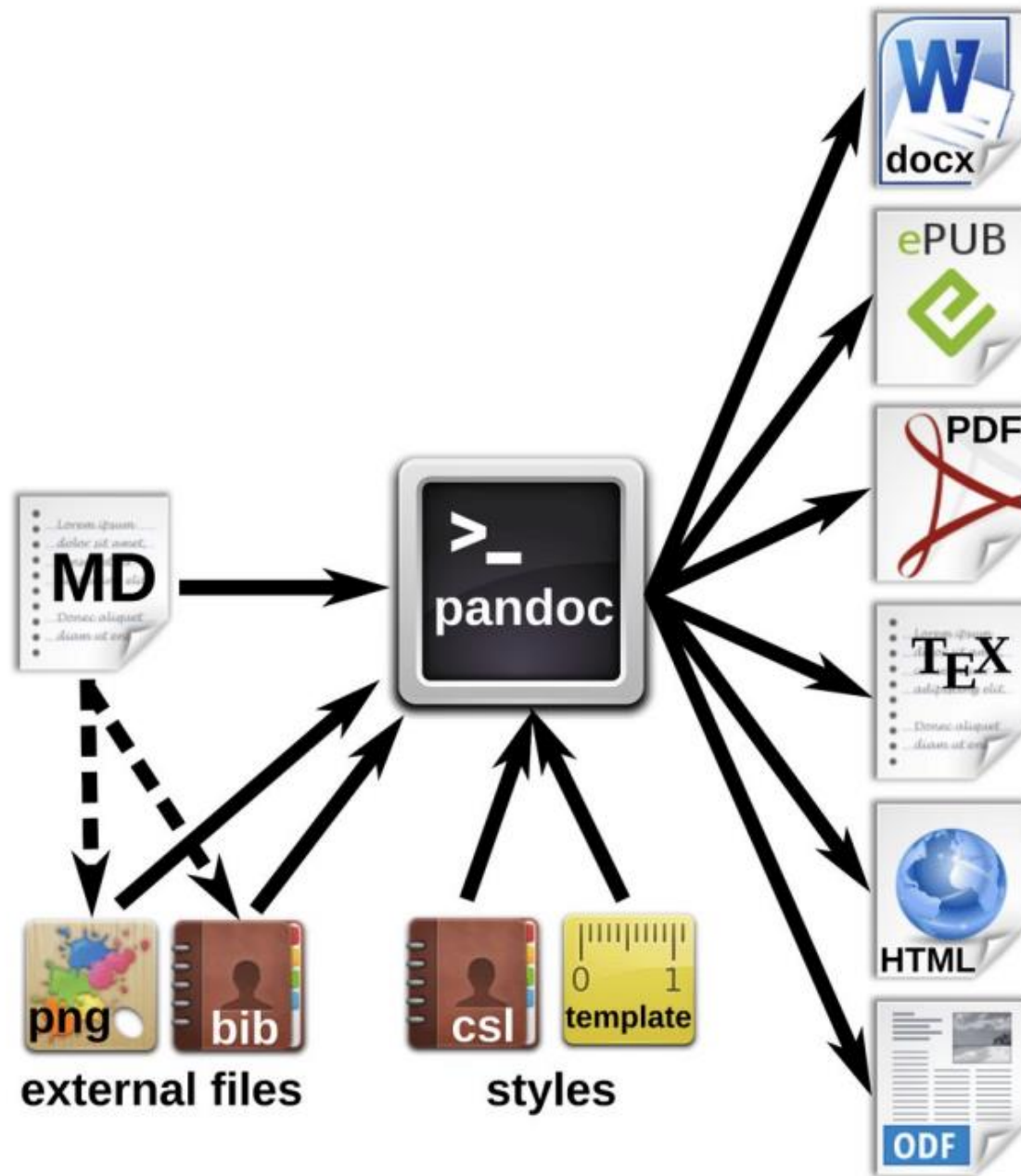
- **Open-source** publishing system (successor to R Markdown)
- Allows for literate programming in Markdown and with integrated tools making reproducibility a default

What is Quarto?

- **Open-source** publishing system (successor to R Markdown)
- Allows for literate programming in Markdown and with integrated tools making reproducibility a default
- Versatility (Choice of editors and output formats)
- Supports R, Python, Julia, ObservableJS.. and potentially more in the future
- Exports to HTML, PDF, Word, books, slides, websites.. over 40 possible formats

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- Supports R, Python, Julia, ObservableJS.. and potentially more in the future
- Exports to HTML, PDF, Word, books, slides, websites.. over 40 possible formats
- Reference management and Cross referencing is possible



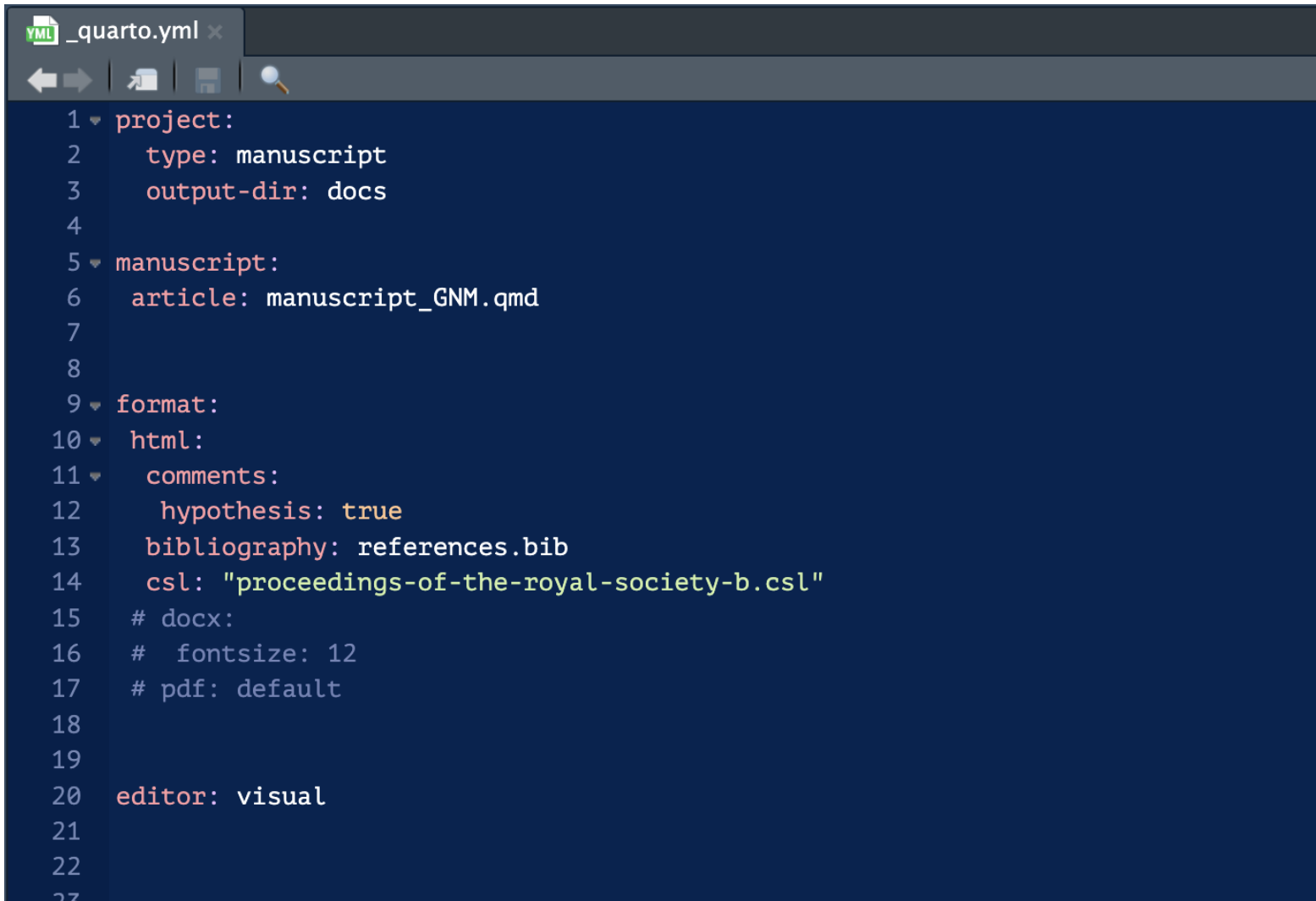
Authoring software options:

- RStudio
- JupyterLab
- VS Code

Components:

- YAML header (metadata)
- Markdown text
- Code chunks + inline code
- Bibliography

YAML header (metadata)

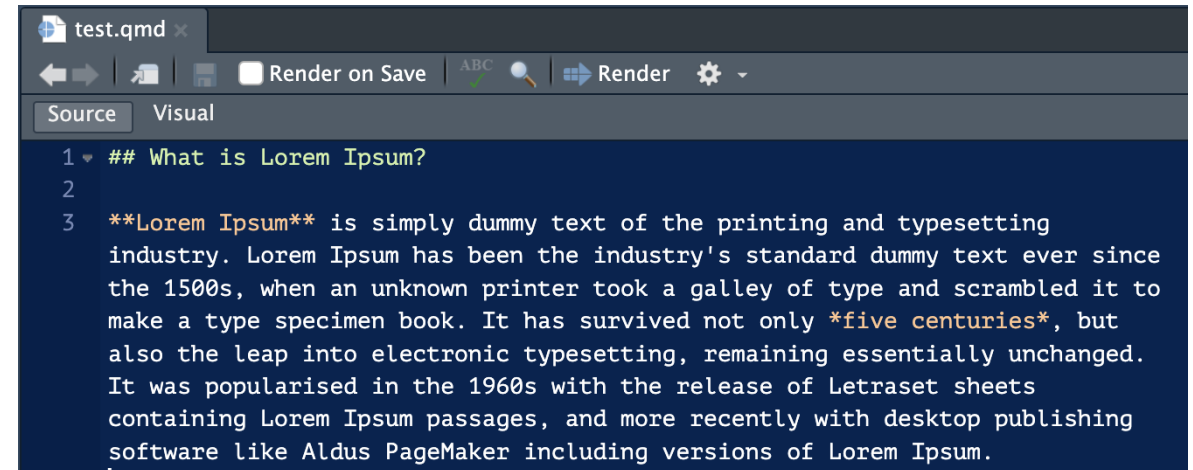


```
1 project:
2   type: manuscript
3   output-dir: docs
4
5 manuscript:
6   article: manuscript_GNM.qmd
7
8
9 format:
10 html:
11   comments:
12     hypothesis: true
13   bibliography: references.bib
14   csl: "proceedings-of-the-royal-society-b.csl"
15   # docx:
16   # fontsize: 12
17   # pdf: default
18
19
20 editor: visual
21
22
23
```


Markdown Text

What is Lorem Ipsum?

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.



```
test.qmd x
Render on Save ABC 🔍 Render ⚙️
Source Visual
1 ## What is Lorem Ipsum?
2
3 Lorem Ipsum is simply dummy text of the printing and typesetting
  industry. Lorem Ipsum has been the industry's standard dummy text ever since
  the 1500s, when an unknown printer took a galley of type and scrambled it to
  make a type specimen book. It has survived not only five centuries, but
  also the leap into electronic typesetting, remaining essentially unchanged.
  It was popularised in the 1960s with the release of Letraset sheets
  containing Lorem Ipsum passages, and more recently with desktop publishing
  software like Aldus PageMaker including versions of Lorem Ipsum.
```

Markdown Text

Markdown Syntax	Output
<code>*italics*</code> and <code>**bold**</code>	<i>italics</i> and bold
<code>superscript^2^</code> / <code>subscript~2~</code>	^{superscript} ² / _{subscript} ₂
<code>~~strikethrough~~</code>	strikethrough
<code>`verbatim code`</code>	<code>verbatim code</code>

Markdown Text

Markdown Syntax

Header 1

Header 1

Header 2

Header 2

Header 3

Header 3

Header 4

Header 4

```
* unordered list
+ sub-item 1
  - sub-sub-item 1
+ sub-item 2
```

- unordered list
 - sub-item 1
 - sub-sub-item 1
 - sub-item 2

```
1. ordered list
2. item 2
  i. sub-item 1
    A. sub-sub-item 1
```

1. ordered list
2. item 2
 - i. sub-item 1
 - a. sub-sub-item 1

```
* item 2

Continued (indent 4 spaces)
```

- item 2
- Continued (indent 4 spaces)

```
term
: definition
```

term
definition

Markdown Text

Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1

Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1

Markdown Text

You can embed [named hyperlinks](https://quarto.org/)

Simply a direct url like <https://quarto.org/>

Links to [other places](#your_link) in the document.

Or embed an image:

![cat](https://upload.wikimedia.org/wikipedia/commons/4/4d/Cat_November_2010-1a.jpg)

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Links to [other places](#) in the document.

Or embed an image:



<https://www.markdownguide.org>

References/Bibliography

A typical BibTex key might look like as follows:

```
@article{10.1371/journal.pone.0300333,  
  doi = {10.1371/journal.pone.0300333},  
  author = {Kambouris, Steven AND Wilkinson, David P. AND Smith, Eden T. AND  
Fidler, Fiona},  
  journal = {PLOS ONE},  
  publisher = {Public Library of Science},  
  title = {Computationally reproducing results from meta-analyses in ecology and  
evolutionary biology using shared code and data},  
  year = {2024},  
  month = {03},  
  volume = {19},  
  url = {https://doi.org/10.1371/journal.pone.0300333},  
  pages = {1-22},  
  number = {3},  
}
```

Referencing workflow (example)

My typical workflow is as follows:

- Find a reference that one needs in your manuscript.

The screenshot shows the PLOS ONE website interface. At the top, the PLOS ONE logo is on the left, and navigation links for 'Publish', 'About', and 'Browse' are on the right. A search bar with a magnifying glass icon and the text 'advanced search' is also present. Below the header, there are icons for 'OPEN ACCESS' and 'PEER-REVIEWED', followed by the text 'RESEARCH ARTICLE'. The main title of the article is 'Computationally reproducing results from meta-analyses in ecology and evolutionary biology using shared code and data'. Below the title, the authors are listed: Steven Kambouris, David P. Wilkinson, Eden T. Smith, and Fiona Fidler. The publication date is March 13, 2024, and the DOI is https://doi.org/10.1371/journal.pone.0300333. On the right side of the article title, there are four yellow boxes showing statistics: 9 Save, 9 Citation, 1,522 View, and 11 Share. Below the title and authors, there is a horizontal navigation bar with tabs for 'Article', 'Authors', 'Metrics', 'Comments', 'Media Coverage', and 'Peer Review'. The 'Article' tab is currently selected. On the left side of the article content, there is a vertical navigation menu with links for 'Abstract', 'Introduction', 'Materials and methods', 'Results', 'Discussion', 'Conclusion', 'Supporting information', 'References', 'Reader Comments', and 'Figures'. The 'Abstract' section is currently visible, showing a paragraph of text. On the right side of the article content, there are buttons for 'Download PDF', 'Print', and 'Share', along with a 'Check for updates' button. At the bottom right, there is an advertisement for the 'Tenth International Congress on Peer Review and Scientific'.

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RESEARCH ARTICLE

Computationally reproducing results from meta-analyses in ecology and evolutionary biology using shared code and data

Steven Kambouris, David P. Wilkinson, Eden T. Smith, Fiona Fidler

Published: March 13, 2024 • https://doi.org/10.1371/journal.pone.0300333

9 Save 9 Citation

1,522 View 11 Share

Article Authors Metrics Comments Media Coverage Peer Review

Abstract

Introduction

Materials and methods

Results

Discussion

Conclusion

Supporting information

References

Reader Comments

Figures

Abstract

Many journals in ecology and evolutionary biology encourage or require authors to make their data and code available alongside articles. In this study we investigated how often this data and code could be used together, when both were available, to computationally reproduce results published in articles. We surveyed the data and code sharing practices of 177 meta-analyses published in ecology and evolutionary biology journals published between 2015–17: 60% of articles shared data only, 1% shared code only, and 15% shared both data and code. In each of the articles which had shared both ($n = 26$), we selected a target result and attempted to reproduce it. Using the shared data and code files, we successfully reproduced the targeted results in 27–73% of the 26 articles, depending on the stringency of the criteria applied for a successful reproduction. The results from this sample of meta-analyses in the 2015–17 literature can provide a benchmark for future meta-research studies gauging the computational reproducibility of published research in ecology and evolutionary biology.

Figures

Download PDF

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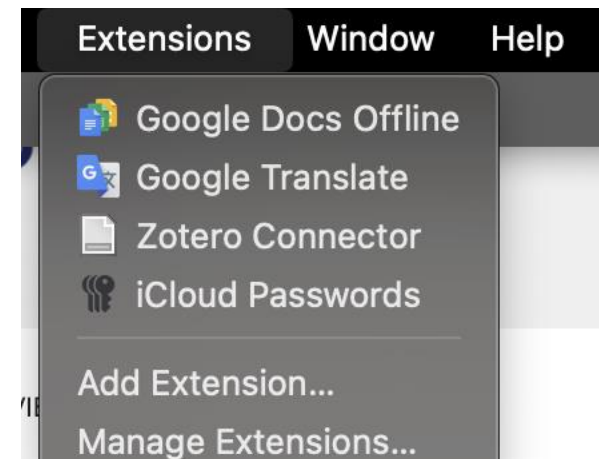
Tenth International Congress on Peer Review and Scientific

Referencing workflow (example)

My typical workflow is as follows:

- Find a reference that one needs in your manuscript.
- Save it to your Zotero library via the browser extension

Zotero automatically updates your .bib file through syncing with Zotero desktop and updating the .bib file.



Referencing workflow (example)

- Cite the references the @ + reference identifier:
 - **In-text citations:**
 - **Bracketed citations:**

Now you can cite the article in the text you are writing @kambouris2024

Now you can cite the article in the text you are writing [kambouris2024]

Now you can cite the article in the text you are writing Kambouris et al. (2024)

Now you can cite the article in the text you are writing (Kambouris et al. 2024)

References

Kambouris, Steven, David P. Wilkinson, Eden T. Smith, and Fiona Fidler. 2024.

"Computationally Reproducing Results from Meta-Analyses in Ecology and Evolutionary Biology Using Shared Code and Data." *PLOS ONE* 19 (3): e0300333.

<https://doi.org/10.1371/journal.pone.0300333>.

References/Bibliography

Switching referencing style according to journals:

- Search your style of preference: <https://www.zotero.org/styles>

Zotero Style Repository

Here you can find [Citation Style Language](#) 1.0.2 citation styles for use with [Zotero](#) and other CSL 1.0.2-compatible software. For more information on using CSL styles with Zotero, see the [Zotero wiki](#).

Style Search

☐ Show only unique styles

Format:

Fields:

7 styles found:

- [ACM SIG Proceedings \("et al." for 15+ authors\)](#) (2023-07-14 21:06:29)
- [ACM SIG Proceedings \("et al." for 3+ authors\)](#) (2025-07-30 08:08:20)
- [Proceedings](#) (2017-06-06 12:27:08)
- [Proceedings of the International Association of Hydrological Sciences](#) (2018-09-09 04:56:26)
- [Proceedings of the National Academy of Sciences of the United States of America](#) (2024-03-27 17:21:14)
- [Proceedings of the National Academy of Sciences, India Section A: Physical Sciences](#) (2014-05-18 01:40:32)
- [Proceedings of the National Academy of Sciences, India Section B: Biological Sciences](#) (2014-05-18 01:40:32)

References/Bibliography

Switching referencing style according to journals:

- Search your style of preference: <https://www.zotero.org/styles>
- Download the csl file into your manuscript folder
- Reference the csl file in your yaml:

```
1 project:
2   type: manuscript
3   output-dir: docs
4
5 manuscript:
6   article: manuscript_GNM.qmd
7
8
9 format:
10 html:
11   comments:
12     hypothesis: true
13     bibliography: references.bib
14     csl: "proceedings-of-the-royal-society-b.csl"
15   # docx:
16   #   fontsize: 12
17   # pdf: default
18
19
20 editor: visual
21
```

Now things become interesting with code

Inline code: Insert results or (quick) calculations into running text, without additional text formatting.

- report and describe results reproducibly
- minimizes errors: no manual copy-pasting results
- automatic updating when data or code changes

Now things become interesting with code

Inline code: Insert results or (quick) calculations into running text, without additional text formatting.

- report and describe results reproducibly
- minimizes errors: no manual copy-pasting results
- automatic updating when data or code changes

```
{r echo=F}  
x <- 5 # radius of a circle
```

For a circle with the radius `r x`, its area is `r pi * x^2`.

For a circle with the radius 5, its area is 78.5398163.

Source your code

Sourcing external files: For example: R and Python scripts, .qmd or .ipynb files.

Why separate scripts from the manuscript?

- minimizes errors: no manual copy-pasting code
- easier code debugging
- making the Quarto document easier to digest
- modular coding = good coding practice

Source your code

```
{r echo=F}  
x <- 5 # radius of a circle
```

For a circle with the radius `r x`, its area is `r pi * x^2`.

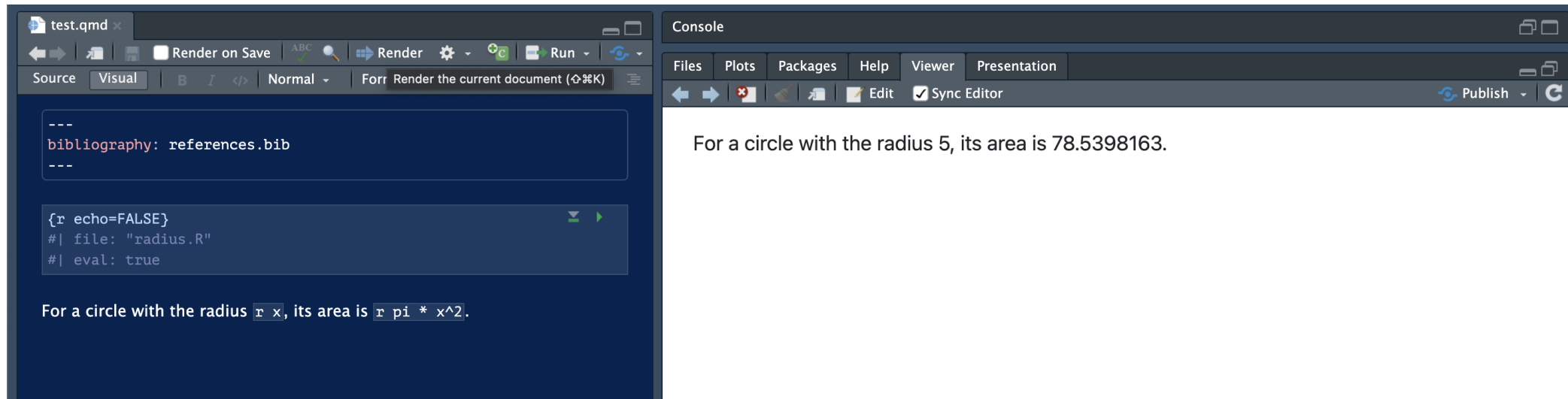
For a circle with the radius 5, its area is 78.5398163.

```
{r echo=FALSE}  
#| file: "radius.R"  
#| eval: true
```

For a circle with the radius `r x`, its area is `r pi * x^2`.

For a circle with the radius 5, its area is 78.5398163.

Render your document



The screenshot displays the Quarto editor interface. The left pane shows the source code of a document named 'test.qmd'. The code includes a bibliography entry and an R code chunk. The right pane shows the rendered output of the document.

Source Code:

```
---  
bibliography: references.bib  
---  
  
{r echo=FALSE}  
#| file: "radius.R"  
#| eval: true
```

Rendered Output:

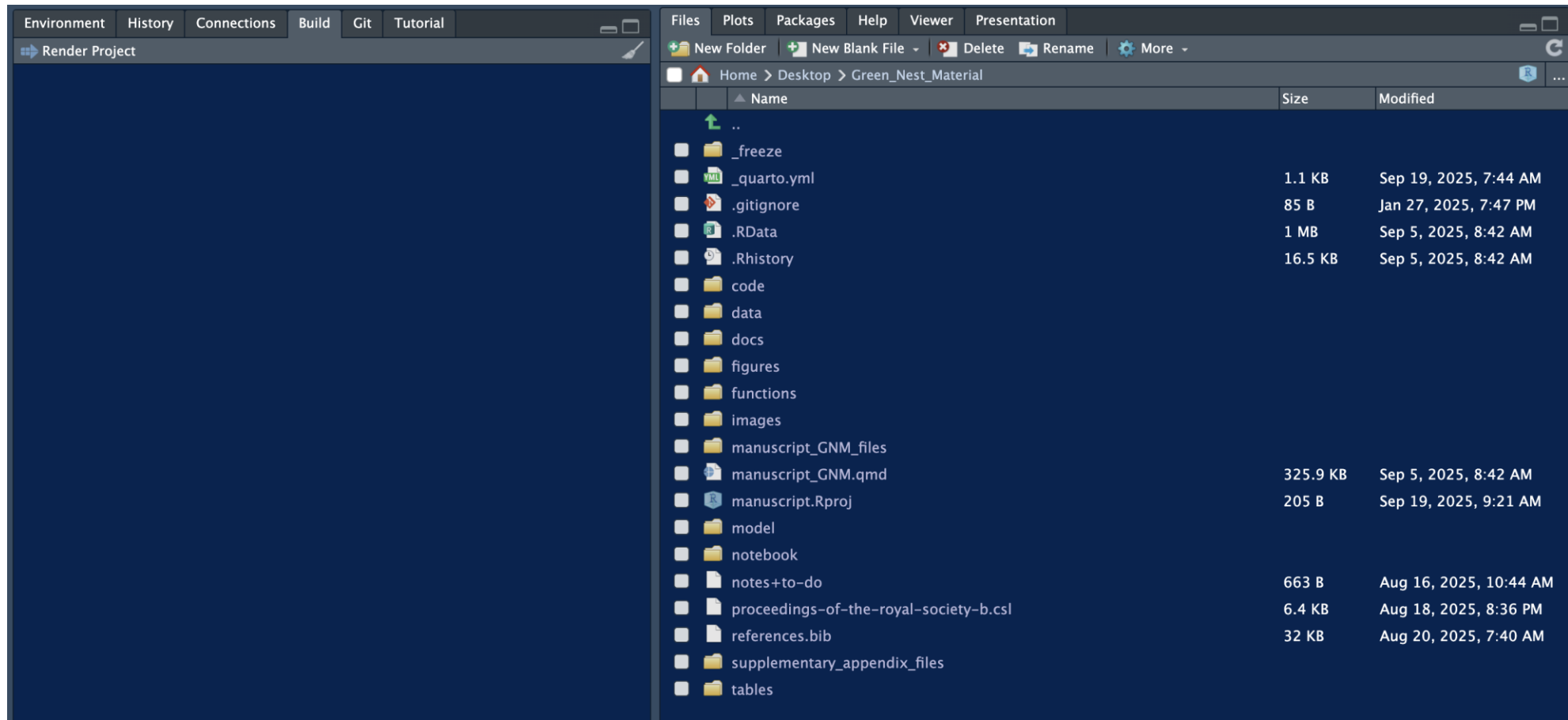
For a circle with the radius r , its area is πr^2 .

Render your document

```
1 library(quarto)
2 quarto_render("my-reproducible-manuscript.qmd") # defaults to html
3 quarto_render("my-reproducible-manuscript.qmd", output_format = "pdf")
```

```
1 quarto render
2 quarto render my-reproducible-manuscript.qmd # defaults to html
3 quarto render my-reproducible-manuscript.qmd --to pdf
```

Render your document



Render your document

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For the few cases for which we could not extract means and variances but inferential statistics such as t and F values, we estimated SMD values ($k = 4$, $n = 2$) using the equations provided by Cooper, Hedges, and Valentine (2019). For dichotomous variables (e.g., number of successful nests, $k = 4$, $n = 2$), we calculated the log odds ratio (lnOR) using the 'escalc' function (measure=OR) from 'metafor' and subsequently converted it to SMD using equations from Cooper, Hedges, and Valentine (2019). Since lnRR cannot be calculated from inferential statistics, dichotomous variables, or traits including zero or negative values ($k = 7$), the number of effect sizes included in the lnRR ($k = 259$, $n = 28$) subset is smaller than that of SMDH ($k = 274$, $n = 28$).

When calculating each effect size's associated sampling variance, we prioritised the number of nests in each group over the number of individuals as the sample size whenever possible to reduce the effect of genetic relatedness non-independence. In cases where the same control or treatment group was used across multiple comparisons, sample sizes were adjusted prior to calculating effect sizes by dividing them by the number of times each group was used in a comparison to account for shared group non-independence. For cases where the authors of an original article reported a statistically non-significant difference between two groups but without providing numerical values ($k = 16$, $n = 2$), we assigned zero as the effect size for that comparison and imputed its sampling variance using the missing-case approach described in Nakagawa, Noble, et al. (2022) for lnRR and using $\frac{n_1 + n_2}{n_1 \times n_2}$ for SMD (Lajeunesse 2013).

Data analysis

We performed all analyses in R v4.5.1 (R Core Team 2025). All multilevel meta-analytical models and meta-regressions were fitted using 'metafor' v.4.8-0 (Viechtbauer 2010). Sampling variances were modelled as variance-covariance matrices assuming a 0.5 correlation (ρ) between sampling variances from the same study (Noble et al. 2017) using the vcalc function from 'metafor' (See Supplementary Appendix [Figure S8](#) for sensitivity analyses using different values of ρ).

Render your document

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[Data cleaning](#)

[Data Preparation](#)

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Results

We obtained 274 effect sizes from a total of 28 studies conducted on 7 species across 17 geographical locations ([Figure1](#)). About 81% of the effect sizes came from three bird species (*Cyanistes caeruleus*, $k = 77$; *Sturnus unicolor*, $k = 60$; *Sturnus vulgaris*, $k = 86$). Our dataset included 2 unpublished studies ($k = 18$), with the remaining studies being published between 1988 and 2024. The mean $\pm$ standard deviation number of effect sizes per study was 9.8 ± 6.4 (median: 8.5 range: 2–24) and the mean effective sample size (i.e., the number of nests/individuals per effect size after accounting for shared-group non-independence) was 32.3 ± 31.1 (median: 22.0, range: 3–210; 31 data points had a sample size of less than 5 nests in each group).

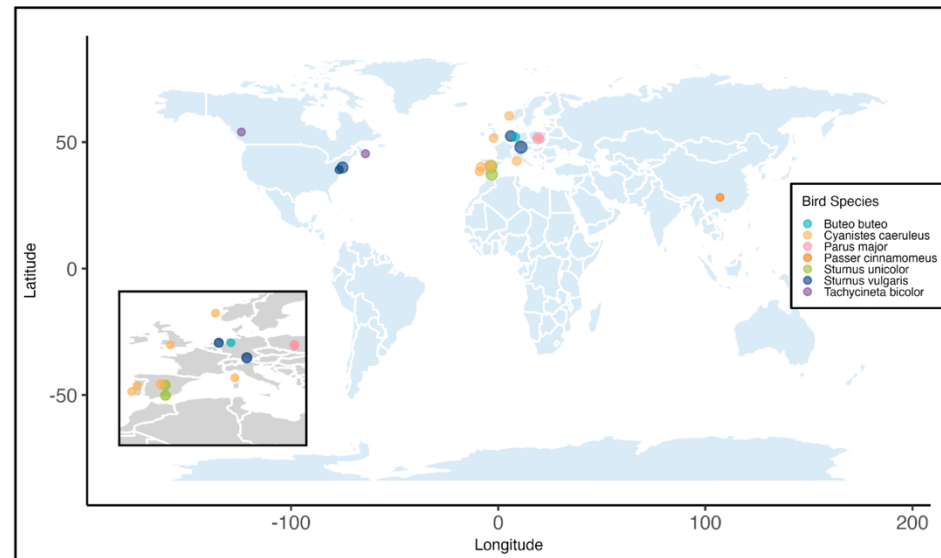


Figure1: **Geographical distribution of the 17 study populations included in a meta-analysis testing the adaptive value of green nest material in birds.** Point size reflects the number of effect sizes coming from the location (range: 3 - 34 effect sizes), and colour denotes species (see legend), with an inset zooming on a dense cluster of data points.

Render your document

Meta-analytic mean [↗](#)

Both lnRR and SMD(H) intercept-only multilevel meta-analyses showed an average increase in fitness estimates in nests with an experimental increase in GNM compared to control nests; however, the overall effect was only statistically significant for SMD(H).

First, lnRR showed an average statistically non-significant increase in fitness outcomes of 1.9% in nests with an experimental increase in GNM ([Figure2](#) A; lnRR = 0.019, 95% CI: [-0.012 to 0.050], 95% PI: [-0.281 to 0.319], p-value = 0.223, k = 238, n = 26) but heterogeneity among effect sizes was high. Total raw heterogeneity (σ^2) was 0.023 ($Q = 2277.7$, $p < 0.001$), the measures of relative ($I^2_{total} = 98.6\%$) and standardised heterogeneity (CVH2_{total} = 63.59 and M2_{total} = 0.98) suggested high heterogeneity across effect sizes.

Second, SMD(H) showed an average statistically significant increase in fitness outcomes in nests with an experimental increase of GNM ([Figure2](#) B; SMD(H) = 0.179, 95% CI: [0.048 to 0.310], 95% PI: [-0.829 to 1.188], p-value = 0.008, k = 253, n = 26), but heterogeneity among effect sizes was moderate to high. Total raw heterogeneity (σ^2) was 0.258 ($Q = 1114.7$, $p < 0.001$), relative heterogeneity was smaller compared to that of lnRR ($I^2_{total} = 66.1\%$) and the standardised heterogeneity metrics (CVH2_{total} = 8.02 and M2_{total} = 0.89) lied between the 50th and 75th percentile of empirically derived estimates from meta-analyses in ecology and evolution ([Yang, Noble, et al. 2024](#)). For lnRR, all three heterogeneity metrics exceeded the 75th

Limitations of Quarto

- Requires R/Python + (for PDF) LaTeX setup
- Learning curve (Markdown/YAML)
- Less familiar to collaborators/editors
- Journal templates sometimes need manual cleanup
- Smaller community (compared to LaTeX or Rmd)

short-term learning curve for long-term gain

Resources

<https://utrechtuniversity.github.io/workshop-reproducible-manuscripts>

<https://lmu-osc.github.io/introduction-to-Quarto/>

<https://mine.quarto.pub/quarto-manuscripts-rmed/#/whats-next>

<https://quarto.org/docs/manuscripts/>

